

# Statistics Lab 52568 — Formula & Concept Sheet

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## Sequence Basics (Lab 1)

**Chargaff's rule:** double-stranded DNA has  $A \approx T$  and  $C \approx G$ .

Consequence: proportion plots of A&T track each other; so do C&G.

**GC content** of a bin of size  $b$ :  $gc_k = (G_k + C_k)/b \in [0, 1]$

**Key graphs:** proportion vs. position (bin each base, plot count/ $b$ ); running median as trend line; scatter plots between base pairs (A vs T, C vs G should be most correlated).

## Coverage & the Poisson Model (Lec 2–3)

$Y_i$  = number of reads starting at base  $i$ , for  $i = 1, \dots, I$ .

$J$  = total number of fragments in the experiment.

**Average coverage:**

$$\lambda = \frac{J}{I} = \frac{\text{total reads}}{\text{total bases}}, \quad \text{avg distance} = \frac{1}{\lambda}$$

**Uniform Poisson — 3 assumptions:**

1. Identical distribution: every fragment has the same  $p_i$
  2. Uniformity:  $p_i = 1/I$  for all  $i$
  3. Independence: fragments land independently
- (1)+(3):  $Y_i \sim \text{Binomial}(J, p_i)$ . Adding (2) + Poisson approx ( $J$  large,  $1/I$  small):

$$Y_i \sim \text{Poisson}(\lambda), \quad \mathbb{E}[Y] = \text{Var}[Y] = \lambda, \quad \text{SD}[Y] = \sqrt{\lambda}$$

**Additive property:**  $X_1, \dots, X_K \sim \text{Pois}(\lambda)$  independent  $\Rightarrow X_1 + \dots + X_K \sim \text{Pois}(K\lambda)$ .

$\Rightarrow$  Bin of size  $b$ :  $Y_k \sim \text{Pois}(\lambda_k)$ ,  $\lambda_k = \sum_{i \in k} p_i \cdot J$ .

**VMR** (Variance-to-Mean Ratio):

$$\text{VMR} = \frac{\text{Var}[Y]}{\mathbb{E}[Y]} = 1 \quad \text{under uniform Poisson}$$

**Key graphs:** histogram of  $Y_i$  vs fitted Poisson bars (log-scale y); coverage vs position with y-axis capped (mark outliers with different symbol); binned coverage full-chromosome plot.

## Overdispersion & Variance Decomposition (Lec 3)

Real data:  $\text{VMR} \approx 14.7$  — uniform model fails. Why?

**Two-level model:**  $Y_k \sim \text{Pois}(\lambda_k)$ , each bin has its own rate.

**Law of Total Variance** (exact identity):

$$\text{Var}(Y) = \underbrace{\mathbb{E}[\text{Var}(Y | \lambda)]}_{\text{avg within-group}} + \underbrace{\text{Var}(\mathbb{E}[Y | \lambda])}_{\text{variance of group means}}$$

Applied here:

$$\text{Var}(Y_k) = \underbrace{\mathbb{E}[\lambda_k]}_{\bar{\lambda} \text{ (Poisson noise)}} + \underbrace{\text{Var}(\lambda_k)}_{\text{rate heterogeneity}}$$

$$\text{VMR} = 1 + \frac{\text{Var}(\lambda_k)}{\bar{\lambda}}$$

$\text{VMR} = 1$  iff all  $\lambda_k$  equal. Spatial correlation of GC prevents  $\text{Var}(\lambda_k)$  from averaging out under binning — rates in nearby bins are similar.

**Robust estimators:** Center: median. Spread: IQR,  $\text{MAD} = \text{med}|x_i - \text{med}(x)|$ . Use instead of mean/SD when data are skewed or have outliers.

**Benjamini & Speed (2012) key findings:** GC bias is **unimodal** (both GC-rich and AT-rich fragments underrepresented; peaks at  $\sim 50\%$  GC). Driven by **full-fragment** GC, not just the read  $\Rightarrow$  PCR is the main cause. Full-fragment model:  $\text{MAD } 9.5$  vs loess  $\text{MAD } 10.8$  on 1 kb bins. GC curves *differ between samples* — always fit per sample.

## Linear Regression for $f(\text{gc})$ (Lec 4)

**Model:**  $Y_k = \beta_0 + \beta_1 \text{gc}_k + \epsilon_k$ ,  $\mathbb{E}[\epsilon_k] = 0$

**OLS estimate:**  $\hat{\beta}_1 = \widehat{\text{Cov}}(\text{gc}, Y) / \widehat{\text{Var}}(\text{gc})$

**Residual plots:** residuals vs. gc and vs. fitted values should be flat. Fan shape  $\Rightarrow$  heteroscedasticity. CIs from **lm** are **wrong** under misspecification.

$$R^2 = 1 - \text{RSS}/\text{TSS}, \quad \text{RMSE} = \sqrt{\frac{1}{K} \sum_k r_k^2}$$

**Global polynomials:** sensitive to outliers — a few extreme points distort the entire fit.

## Spline Regression (Lec 4)

Key idea: divide the predictor into regions at **knots**  $t_1, \dots, t_m$ ; fit a polynomial in each piece; glue with smoothness constraints.

**Progression:** piecewise constant  $\rightarrow$  piecewise linear (discontinuous)  $\rightarrow$  continuous linear spline  $\rightarrow$  cubic spline (continuous 1st & 2nd derivatives).

**Cubic spline basis:**  $\hat{y} = b_0 + b_1x + b_2x^2 + b_3x^3 + \sum_j c_j(x - t_j)_+^3$ , where  $(u)_+ = \max(u, 0)$ .

**B-spline** `bs(x,knots,degree=3)`: locally supported basis, same fit. More knots = more flexibility.

**Heteroscedasticity check:**

$$\text{SD ratio} = \frac{\text{SD}(\text{resid}, \text{gc} > \text{Q3})}{\text{SD}(\text{resid}, \text{gc} < \text{Q1})} \approx 2.25$$

SD ratio  $\gg 1 \Rightarrow$  variance grows with gc/coverage.

## Poisson GLM & Quasi-Poisson (Lab 5)

**Poisson GLM:**  $\log \mathbb{E}[Y_k] = \beta_0 + \beta_1 \text{gc}_k$  (log link, counts).

Assumes  $\text{Var}(Y) = \mathbb{E}[Y]$  — still violated ( $\phi \approx 14$ ).

**Quasi-Poisson:** same mean,  $\text{Var}(Y) = \phi \mathbb{E}[Y]$ ,  $\phi$  estimated from data. More honest uncertainty.

**Comparing models:**  $R^2$  for lm; McFadden pseudo- $R^2$  for GLM;  $\tilde{R}^2 = 1 - \frac{\log L(\text{model})}{\log L(\text{null})}$ .

Spearman  $r$  is rank-based — robust to outliers, but misleading when there is no real signal (centromere paradox: Spearman  $\approx 0.95$  despite  $R^2 \approx 0$ ).

**Bin size tradeoff:** larger bins  $\downarrow$  Poisson noise,  $\uparrow$  GC signal blurring. Statistical elbow  $\sim 5$  kb. Biological constraint: genes 10–100 kb.

## Copy Number Estimation (Labs 6–8, Yuval's notes)

**Model:**  $Y_k \sim \text{Pois}(\lambda_k)$ ,

$$\lambda_k = a_k \cdot f(\text{gc}_k) \cdot \epsilon_k$$

$a_k \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots\}$  = copy number.  $f(\text{gc}_k)$  = expected coverage at copy number 1.  $\epsilon_k$  = residual technical noise,  $\mathbb{E}[\epsilon_k] = 1$ .

Identifiability:  $\text{med}_k\{a_k\} = 1$ ,  $\mathbb{E}_k[f(\text{gc}_k)] = 1$ .

**Single-sample plug-in estimator**  $\hat{a}_k^{(1)}$ :

$$\hat{a}_k^{(1)} = \frac{Y_k}{\hat{f}(\text{gc}_k)} \cdot \hat{M}, \quad \hat{M} = \frac{1}{\text{med}_k\left\{\frac{Y_k}{\hat{f}(\text{gc}_k)}\right\}}$$

**Stabilized estimator** (reduces variance, increases bias for  $a_k \neq 1$ ):

$$\hat{a}_k^{(1*)} = \frac{Y_k + c}{\hat{f}(\text{gc}_k) + c}$$

**Variance of  $\hat{a}_k^{(1)}$ :**

$$\text{Var}(\hat{a}_k) \approx \underbrace{\frac{1}{f(\text{gc}_k)}}_{\text{Poisson} \approx 16\%} + \underbrace{\frac{\sigma^2}{f(\text{gc}_k)^2}}_{\epsilon \text{ noise} \approx 84\%}$$

**Two-sample model** (Yuval's note 2): decompose noise into shared  $\gamma_k$  and sample-specific  $\delta_k^T, \delta_k^N$ :

$$Y_k^T = a_k \cdot f^T(\text{gc}_k) \cdot \gamma_k \cdot \delta_k^T + \eta^T, \quad Y_k^N = f^N(\text{gc}_k) \cdot \gamma_k \cdot \delta_k^N + \eta^N$$

**Two-sample estimator:**

$$\hat{a}_k^{(2)} = \frac{Y_k^T / \hat{f}^T(\text{gc}_k) + c}{Y_k^N / \hat{f}^N(\text{gc}_k) + c} \approx a_k \cdot \frac{\delta_k^T}{\delta_k^N}$$

Shared noise  $\gamma_k$  cancels. Residual noise is ratio of sample-specific terms only — better than single-sample if  $\gamma_k$  dominates.

**Fit separate GC spline per sample.** Ratio also cancels germline CNVs.

**Noise models:** Additive:  $\lambda_k = a_k f + \epsilon$  (offset independent of signal). Multiplicative:  $\lambda_k = a_k f \cdot \eta$ ,  $\eta \sim N(1, \gamma^2)$  — error scales with signal; PCR amplifies proportionally  $\Rightarrow$  more plausible.

**Key diagnostic:** plot corrected ratio vs position — flat at 1 under  $H_0$ . Y-axis must be corrected ratio, not raw reads.

**Aggregative model (Lab 7):** fit at 1 kb, sum sub-cell predictions to predict 10 kb bins. Preserves local GC structure.

## Hypothesis Testing & Multiple Testing (Lab 9)

**Statistic:**  $S_k = \log_2(\hat{a}_k^{2\text{-sample}})$ . Under  $H_0$ :  $S_k \approx 0$ .

**Negative controls:** locally stable bins (window median  $\leq \log_2 1.10$ , SD  $\leq \log_2 1.15$ ), filtered on the *ratio* so germline CNVs pass correctly.

**Empirical p-value:**  $p_k = |\{j : |S_j| \geq |S_k|\}| / |\text{ctrl}|$ . No distributional assumption.

**Calibration:** P-P plot of held-out CN=1 bin p-values should track the uniform diagonal.

**Bonferroni** (FWER):  $\alpha' = \alpha/m$ . Too conservative ( $m \sim 10^5$ ).

**Benjamini–Hochberg** (FDR): reject  $H_{(k)}$  for  $k \leq k^* = \max\{k : p_{(k)} \leq k\alpha/m\}$ .

Controls  $\mathbb{E}[\text{FP}/\text{sig}] \leq \alpha$ . Preferred: tolerates a small known FP fraction for much greater power.

## Key Numbers

|                      |                                                              |
|----------------------|--------------------------------------------------------------|
| $\lambda$ (50–60 Mb) | $\approx 0.036/\text{base}$                                  |
| VMR (binned)         | $\approx 14.7$ ; GC-cov $r \approx 0.84$                     |
| Spline $R^2$ ceiling | $\approx 0.75$ ; SD ratio $\approx 2.25$                     |
| $\phi$ quasi-Pois    | $\approx 14$ ; $R^2$ gain G+C $\approx 0.17$                 |
| CN MAE               | $\approx 0.15$ ; Var: 16% Pois / 84% $\epsilon$              |
| $r$ tumor/healthy    | $\approx 0.10$ ; elbow $\sim 5$ kb                           |
| Centromere gap       | $\sim 120\text{--}130$ Mb; GC peak $\sim 53\text{--}55.5$ Mb |
| Paper MAD            | fragment 9.5 vs loess 10.8 (1 kb bins)                       |